

Map coloring problem for Telangana districts

Objective: Color the districts of Telangana such that adjacent districts do not have the same color.

```
import matplotlib.pyplot as plt
import networkx as nx

districts = [
    "Hyderabad",
    "Rangareddy",
    "Medchal",
    "Sangareddy",
    "Medak",
    "Siddipet",
    "Yadadri",
    "Nalgonda",
    "Mahabubnagar",
    "Nagarkurnool",
    "Wanaparthi",
    "Jogulamba",
    "Vikarabad",
    "Narayanpet"
]

graph = {
    "Hyderabad": ["Rangareddy", "Medchal"],
    "Rangareddy": [
        "Hyderabad", "Medchal", "Sangareddy",
        "Vikarabad", "Mahabubnagar", "Nalgonda"
    ],
    "Medchal": ["Hyderabad", "Rangareddy", "Sangareddy", "Medak"],
    "Sangareddy": [
        "Medchal", "Rangareddy", "Vikarabad", "Medak"
    ],
    "Medak": ["Sangareddy", "Medchal", "Siddipet"],
    "Siddipet": ["Medak", "Yadadri"],
    "Yadadri": ["Siddipet", "Nalgonda"],
    "Nalgonda": ["Yadadri", "Rangareddy", "Mahabubnagar"],
    "Mahabubnagar": [
        "Rangareddy", "Nalgonda", "Nagarkurnool",
        "Wanaparthi", "Narayanpet"
    ],
    "Nagarkurnool": ["Mahabubnagar", "Wanaparthi"],
    "Wanaparthi": ["Mahabubnagar", "Nagarkurnool", "Jogulamba"],
    "Jogulamba": ["Wanaparthi", "Narayanpet"],
    "Vikarabad": ["Rangareddy", "Sangareddy", "Narayanpet"],
    "Narayanpet": ["Vikarabad", "Mahabubnagar", "Jogulamba"]
}

colors = ["Red", "Green", "Blue", "Yellow"]

color_assignment = {}

def is_valid(district, color):
    for neighbor in graph[district]:
        if color_assignment.get(neighbor) == color:
            return False
    return True

def map_coloring(index=0):
```

```

    if index == len(districts):
        return True

    district = districts[index]

    for color in colors:

        if is_valid(district, color):

            color_assignment[district] = color

            if map_coloring(index + 1):
                return True

            del color_assignment[district]

    return False

if map_coloring():

    print("Map Coloring Solution:\n")

    for district, color in color_assignment.items():
        print(district, ":", color)

    print("\nAll adjacent districts in the modeled graph have different colors.")

    G = nx.Graph()

    G.add_nodes_from(districts)

    for district in graph:

        for neighbor in graph[district]:
            G.add_edge(district, neighbor)

    node_colors = [
        color_assignment[district].lower()
        for district in G.nodes()
    ]

    pos = nx.spring_layout(G, seed=42, k=1.5)

    plt.figure(figsize=(10, 5))

    nx.draw_networkx_edges(
        G,
        pos,
        edge_color="gray",
        width=0.5
    )

    nx.draw_networkx_nodes(
        G,
        pos,
        node_color=node_colors,
        node_size=2800,
        edgecolors="black",
        linewidths=0.5
    )

    nx.draw_networkx_labels(
        G,
        pos,
        font_size=9,
        font_weight="bold",
        font_color="white"
    )

    plt.title(
        "Telangana District Map Coloring Using Backtracking",
        fontsize=10,
        fontweight="bold"
    )

    plt.axis("off")
    plt.tight_layout()
    plt.show()

else:

```

```
print("No valid coloring found")
```

Map Coloring Solution:

Hyderabad : Red
Rangareddy : Green
Medchal : Blue
Sangareddy : Red
Medak : Green
Siddipet : Red
Yadadri : Green
Nalgonda : Red
Mahabubnagar : Blue
Nagarkurnool : Red
Wanaparthi : Green
Jogulamba : Red
Vikarabad : Blue
Narayanpet : Green

All adjacent districts in the modeled graph have different colors.

Telangana District Map Coloring Using Backtracking

